

AI Research Agent

Project Overview

The AI Research Agent is a simple intelligent system designed to answer research questions by combining a large language model with web search and automated decision-making.

Instead of only generating answers from the model's internal knowledge, the agent can search the web, read webpages, collect information from multiple sources, and then generate a structured final answer.

The project uses Groq, LangChain, LangGraph, LangSmith, DuckDuckGo Search, and BeautifulSoup.

Main Technologies Used

Groq

Groq is used to run the large language model.

The model understands the user's question, decides whether web research is needed, analyzes the collected information, and produces the final answer.

In this project, Groq is connected through LangChain.

LangChain

LangChain is used to connect the language model with external tools.

It manages:

- Groq model integration
- User and AI messages
- Web search tools
- Webpage-reading tools
- Tool calling

LangChain gives the AI access to functions outside the language model.

LangGraph

LangGraph controls the workflow of the AI agent.

The basic workflow is:

User Question → LLM → Tool Needed? → Web Search / Webpage Reading → LLM → Final Answer

If the model needs more information, LangGraph sends it back to the tools and repeats the process.

This allows the system to work as an agent rather than a simple chatbot.

LangSmith

LangSmith is used for monitoring, debugging, and tracing the agent.

It allows the developer to see:

- Which prompts were sent
- Which tools were called
- What results the tools returned
- How the agent moved through the workflow
- Errors and execution time

This makes it easier to understand and improve the system.

How the System Works

The user enters a research question.

The Groq language model analyzes the question and decides whether external information is required.

If research is needed, the agent calls the `web_search` tool.

The search tool returns relevant websites.

The model can then call the `web_readpage` tool to read selected webpages.

The model analyzes the collected information and may perform additional searches if necessary.

When enough information has been collected, the agent produces the final research answer.

LangGraph controls this process, while LangSmith records the execution for debugging and monitoring.

Main Features

- AI-powered question answering
- Automatic web searching
- Automatic webpage reading
- Tool selection by the AI

- Multi-step research workflow
- Source-based research
- LangSmith tracing and debugging
- Free tools suitable for learning and portfolio projects

Conclusion

The AI Research Agent demonstrates how modern AI agents can combine language models with external tools.

LangChain connects the model and tools, LangGraph controls the research workflow, LangSmith monitors the system, and Groq provides the language model.

The result is a research assistant that can search for information, analyze sources, and generate useful answers instead of relying only on the model's existing knowledge.